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VOLTAGE AND ION DEPENDENCES OF THE SLOW CURRENTS WHICH MEDIATE BURSTING IN *APLYSIA* NEURONE R15

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SUMMARY

1. The previous paper described a slow depolarizing tail current, I_D, and a slow hyperpolarizing tail current, I_H, that are activated by action potentials and by brief depolarizing pulses in *Aplysia* neurone R15. I_D and I_H are necessary for the generation of bursting pace-maker activity in this cell. In this paper, the voltage and ion dependence of I_D and I_H are studied in an effort to determine the charge carriers for the two currents.

2. When the slow currents are activated by brief depolarizing pulses delivered under voltage clamp in normal medium, an increase in the size of the pulse of 5–10 mV is usually sufficient to bring about full activation of I_D. The apparent threshold in normal medium is approximately -20 mV. In medium in which K^+ channels are blocked, full activation of an inward tail current that resembles I_D requires increasing the pulse amplitude by only 1–2 mV. In contrast, I_H is activated in a graded fashion over a 40 mV range of pulse amplitudes.

3. After activating the currents with action potentials or with supramaximal pulses, I_D remains an inward current and I_H an outward current over a range of membrane potentials spanning -20 to -120 mV.

4. In normal medium, I_D is dependent on both extracellular Na^+ concentration ($[Na^+]_o$) and extracellular Ca^{2+} concentration ($[Ca^{2+}]_o$). When K^+ channels are blocked, I_D can be supported by either $[Na^+]_o$ or $[Ca^{2+}]_o$. I_H depends only on $[Ca^{2+}]_o$ as long as $[Na^+]_o$ is at least 50 mM.

5. Neither I_D nor I_H is decreased by decreasing the K^+ gradient or by application of K^+ channel blockers. These treatments increase somewhat the apparent amplitude of I_D, probably by unmasking it from the large K^+ tail current that follows the depolarizing pulse.

6. A direct comparison in the same cell of the tetraethylammonium sensitivity of I_H and of the Ca^{2+}-activated K^+ current demonstrates that these two currents flow through separate and distinct populations of channels.

7. We conclude that in R15, I_D arises in response to the triggering of an axonal action potential which in turn, through an as yet unknown mechanism, causes an increased influx of Na^+ and/or Ca^{2+}.

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8. We conclude that the apparent outward current I_H, which is responsible for the interburst hyperpolarization in a normally bursting R15, in fact arises from a *decrease* in a resting inward Ca^{2+} current, possibly as the result of Ca^{2+}-induced inactivation of Ca^{2+} channels.

INTRODUCTION

The previous paper (Adams, 1985) described the kinetics of two slow currents in *Aplysia* neurone R15 that are activated by action potentials. One of these currents, I_D, is an inward, or depolarizing current that reaches a peak 300–500 ms after the action potential. It produces the depolarizing after-potential that follows action potentials in this cell and is responsible also for the grouping together of action potentials into bursts. The second current, I_H, is an outward, or hyperpolarizing current that is smaller and slower than I_D. It reaches a peak in 2–10 s and is still present for many tens of seconds following the action potential. Because of the long duration of I_H, the incremental contributions from each action potential in the burst summate, until the summated amplitude of I_H is large enough to cause the cell to hyperpolarize, thereby bringing the burst to an end.

In this paper, we investigate the voltage and ion dependences of I_D and I_H in an effort to determine the charge carriers for the two currents. The results show that I_D is activated by a threshold process, that it remains an inward current between -20 and -120 mV, and that it requires both Na^+ and Ca^{2+} in the bathing medium. I_H is activated in graded fashion, it remains an outward current between -20 and -120 mV, and it is dependent only on extracellular Ca^{2+} concentration ($[Ca^{2+}]_o$). Neither I_D nor I_H is decreased by decreases in the K^+ gradient or by K^+ channel blockers. In particular, although it has been proposed (Gorman, Hermann & Thomas, 1982) that the interburst hyperpolarization in neurone R15 is mediated by a Ca^{2+}-activated K^+ current, our evidence indicates that I_H, while dependent upon extracellular Ca^{2+} and Ca^{2+} entry into the cell, is not carried by K^+. A similar conclusion has been drawn by Kramer & Zucker (1985*b*) for bursting neurones from the left upper quadrant of the *Aplysia* abdominal ganglion. Furthermore, a direct comparison of the tetraethylammonium (TEA) sensitivity of I_H and the Ca^{2+}-activated K^+ current demonstrates that they flow through different channels. Our results are most consistent with the idea that I_H (and hence the interburst hyperpolarization) results from the inactivation of a persistent inward Ca^{2+} current. Portions of this work have been published previously in abstract form (Adams & Levitan, 1981, 1982).

METHODS

The procedures were essentially the same as those described in the previous paper (Adams, 1985). The one exception is that the measurements in this paper were made over a wide range of membrane potentials. When dealing with normally bursting cells (Fig. 1), the voltage-clamp holding potential was pre-set to the measurement potential and the clamp was activated automatically when the membrane potential passed through -45 mV with positive slope. For measurements made with simulated action potentials (Figs. 2–16), the command pulse wave forms are described in the Results section.

RESULTS

Voltage dependence of I_D and I_H

Normally bursting cells. In order to measure the voltage dependence of I_D and I_H, the cell was allowed to burst freely and the voltage clamp was turned on to clamp the cell at different measurement potentials after (a) the first, (b) the second or (c) the seventeenth action potential in a burst. The maximum inward difference current between measurements (b) and (a) was taken as a measure of I_D; the maximum

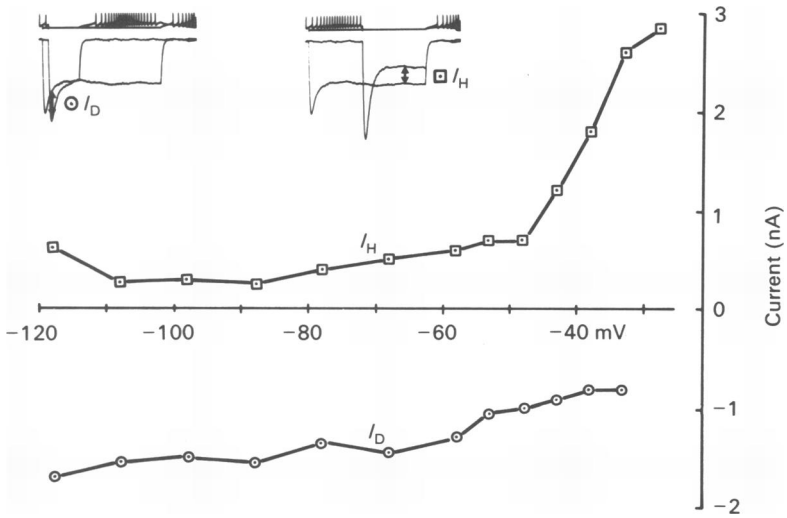


Fig. 1. Voltage dependence of I_D and I_H in a normally bursting cell. I_D (⊙) was measured as the difference current between the second and first action potentials and I_H (⊡) as the difference current between the seventeenth and first action potentials in the burst. The current traces were positioned in real time, by triggering them on the first action potential in the burst, before subtracting to obtain the difference currents (see Adams, 1985). Inset: measurement of I_D and I_H at −68 mV.

outward difference current between measurements (c) and (a) was taken as a measure of the summated I_H from sixteen action potentials. To minimize the effects of the continually changing background current (as mentioned in the previous paper), the current traces were positioned in real time before subtracting them to get the difference current. The results are shown in Fig. 1. I_D remains an inward current over the voltage range −33 to −118 mV, increasing somewhat with hyperpolarization. I_H decreases rapidly with hyperpolarization between −28 and −48 mV and then decreases more slowly with further hyperpolarization. However, it remains an outward current over the entire voltage range from −28 to −118 mV.

One drawback of the above method is that voltage clamps that take the membrane potential far out of the range normally encountered during bursting may result in activation or inactivation of currents which are not directly associated with bursting, but which, by changing, can affect the measurements of the burst currents. To minimize this problem, further measurements were made by simulating action potentials with brief depolarizing pulses.

Simulated action potentials. The use of simulated action potentials under voltage clamp allows a more direct measurement of the voltage dependence of the slow currents. Two types of voltage dependence are of interest: (1) the membrane potential that must be reached during the action potential or depolarizing pulse in order to activate the slow currents and (2) once activated, the way in which the slow currents

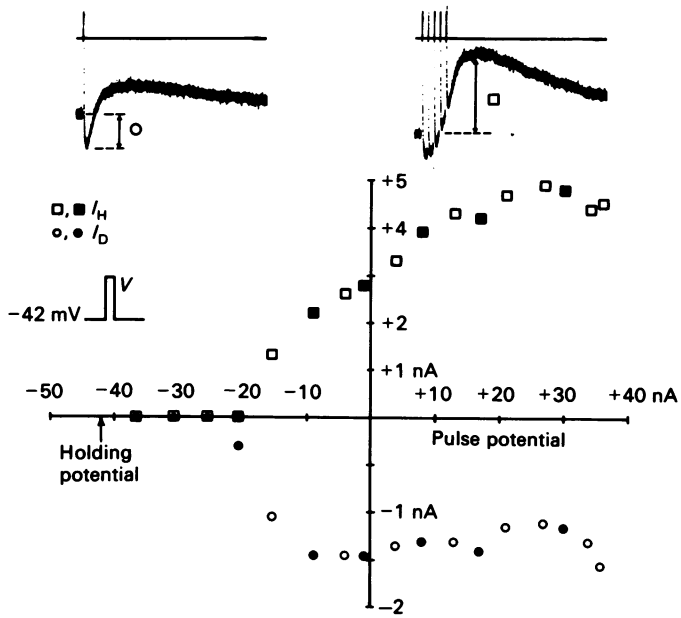


Fig. 2. Activation curves for I_D and I_H. The cell was held at −42 mV, depolarized to potential 'V' for 10 ms, and returned to −42 mV to measure I_D (○, ●) and I_H (□, ■). A single depolarizing pulse was used to activate I_D and a train of five pulses at 0·5 s intervals to activate I_H. The open symbols (○, □) represent data from an 'ascending series' of measurements with increasingly larger pulses, the filled symbols (●, ■) from a subsequent 'descending series'. Insets: sample voltage and current traces for depolarizing pulses to −1 mV. The arrows indicate the times at which I_D and I_H were measured.

depend on the membrane potential at which they are measured. In order to minimize cross-contamination between measurement of the two currents, I_D was measured in response to activation by a single depolarizing pulse, while I_H was measured in response to activation by five pulses. Under most conditions, the peak of I_D is several times larger than that of I_H and, moreover, the inward current peak occurs well before the outward current peak (cf. Adams, 1985, Fig. 6). On the other hand, the outward current summates nearly linearly over several depolarizing pulses (cf. Adams, 1985, Fig. 7) while the inward current summates only over two or three pulses. Thus the outward current activated by several depolarizing pulses provides a less-contaminated measure than that activated by a single pulse. In practice, stimuli were generated every 90 s, alternating between a single depolarizing pulse to measure I_D and a train of five depolarizing pulses to measure I_H.

Activation of I_D and I_H was found to depend differently on the membrane potential reached during the depolarizing pulse (Fig. 2). Both slow currents required the

membrane potential to reach approximately −20 mV for any activation. Full activation of the inward current was achieved when the depolarizing pulse reached −10 mV, a range of only 10 mV between minimal and maximal activation. In contrast, the outward current continued to grow with increasing pulse amplitude and reached a maximum only for pulses which depolarized the cell to +20 mV, a range of approximately 40 mV between minimal and maximal activation.

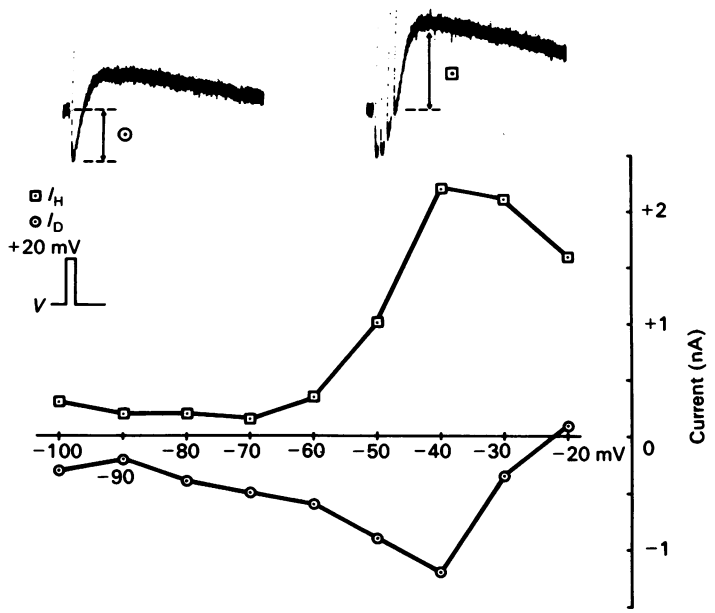


Fig. 3. Voltage dependence of I_D and I_H (method I). The cell was held at voltage 'V', depolarized to +20 mV for 10 ms, and returned to 'V' to measure I_D (⊙) and I_H (⊡). Insets: measurement of I_D and I_H at −40 mV.

The activation curve for I_H was similar in all cells studied, with activation beginning at approximately −20 mV and requiring a further 30–50 mV to reach full amplitude. However, much cell-to-cell variation was encountered in the activation of I_D. Minimal activation was usually found near −20 mV, but the voltage range between minimal and maximal activation varied widely. In one cell, the activation curve for I_D was as broad as that for I_H. In several cells, full activation of I_D was achieved over a range of only 2–3 mV. The activation curve for I_D illustrated in Fig. 2 represents a median between these extremes. Several such cells were encountered, but because of the large variation, they cannot be considered truly 'typical'.

Two different schemes were used to measure the voltage dependence of I_D and I_H once they had been fully activated by depolarizing pulses. In the simpler of the two (Fig. 3), the membrane potential was held at various potentials, pulsed to +20 mV for 10 ms, and returned to the holding potential for measurement of I_D and I_H. Both currents were maximal at a holding and measuring potential of −40 mV. The currents decreased in amplitude at more positive potentials, possibly because they

were already partially activated or inactivated at these potentials. As the holding potential was made more negative the amplitude of both currents became smaller, but I_D remained an inward current and I_H an outward current to at least -100 mV. There appeared to be a small increase in the size of the currents at very negative potentials.

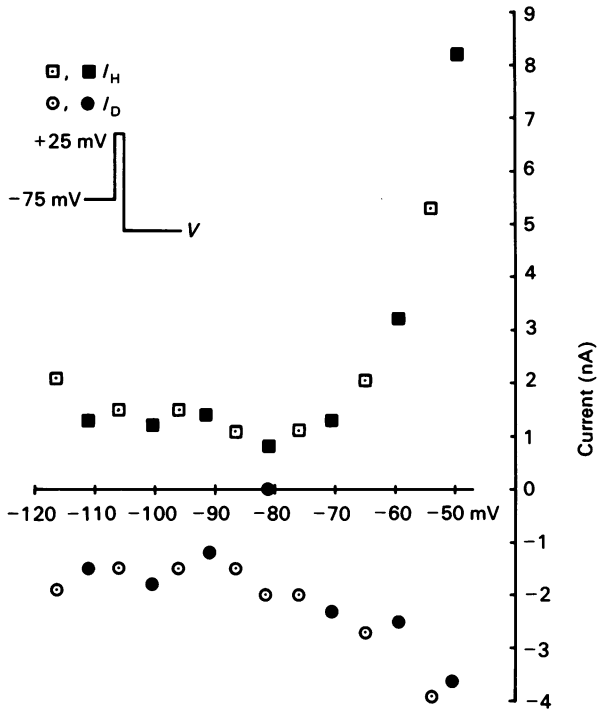


Fig. 4. Voltage dependence of I_D and I_H (method II). The cell was held at -75 mV, depolarized to $+25$ mV for 10 ms, and then stepped to voltage 'V'. The procedure was repeated without the depolarizing pulse and I_D (○, ●) and I_H (□, ■) were measured as difference currents. The open symbols (○, □) represent data from an 'ascending series' of measurements with increasingly more positive measuring potentials, the filled symbols (●, ■) from a subsequent 'descending series'.

In the second scheme (Fig. 4), the holding potential was kept constant at -75 mV. The membrane potential was pulsed to $+25$ mV for 10 ms and then stepped immediately to the measurement potential. The procedure was then repeated without the depolarizing pulse and the difference current during the measurement phase was computed to obtain I_D and I_H. The results are similar to those in Fig. 3, except that it proved impossible to step to a voltage more positive than -50 mV without triggering axonal action potentials. The amplitudes of both I_D and I_H were largest at the most depolarized potentials, but while I_D decreased gradually with hyperpolarization, I_H fell rather abruptly until the membrane potential was -60 to -80 mV and then continued falling more slowly, with perhaps a slight increase at very hyperpolarized potentials.

The voltage dependences of the currents, elicited by action potentials (Fig. 1) or by depolarizing pulses (Figs. 3 and 4), show a number of similarities. We believe that

any differences arise in part from differences between individual cells and in part from the different techniques by which the data were obtained. Among the similarities, one must note that, independent of the method of measurement, I_D remains an inward current and I_H an outward current over the entire range of membrane potentials. In addition, the voltage dependence of I_D is rather 'soft', increasing (Fig. 1) or decreasing (Figs. 3 and 4) perhaps 2-fold with hyperpolarization. In contrast, I_H decreases some 6–8-fold with hyperpolarization, rather rapidly over the range of membrane potentials that the cell might encounter during normal bursting, and then more slowly for more hyperpolarized potentials. The transition in the current–voltage (I–V) curves for I_H, from strong voltage dependence to a more gradual voltage dependence, occurs generally between -50 and -70 mV, and points out the error that would occur if the I–V curves in the more depolarized range were simply extrapolated to obtain a 'reversal potential'. Such extrapolations would predict reversal potentials between -55 and -75 mV, when in fact I_H does not reverse even at potentials as negative as -118 mV.

Ion dependence of I_D and I_H

The effects of ion replacements and of ion channel blockers were measured in similar fashion. The cell was voltage clamped at or near -40 mV. Brief depolarizing pulses were presented, singly to measure I_D, and alternately in a group of five at 0·5 s intervals to measure I_H. A 90 s interval was allowed for recovery between each measurement. When the peak values of I_D and I_H had stabilized, the perfusion medium was changed and perfusion with the new medium was continued until stable measurements were again obtained (usually 15–50 min). The preparation was then returned to normal medium to follow the recovery.

Na^+. Decreasing extracellular Na^+ decreased the amplitude of I_D without affecting I_H, as long as the extracellular Na^+ concentration ($[Na^+]_o$) remained greater than 50 mM. Fig. 5 (top) portrays successive measurements of I_D and I_H as $[Na^+]_o$ was decreased from 460 mM (normal) to 100 mM and then returned to 460 mM. I_D decreased within 15 min to 20% of its value in normal medium. Upon return to normal medium I_D recovered within 10 min. In contrast, I_H was unaffected by this change in $[Na^+]_o$.

Although the amplitude of I_H was unchanged by partial reduction of $[Na^+]_o$, it could be blocked reversibly by total removal of Na^+ from the bathing medium. This is illustrated in Fig. 5 (bottom). The block of I_H was quite rapid, nearly as rapid as that of I_D, and both returned to their control amplitudes, following an 'overshoot', when the normal concentration of Na^+ was restored to the bathing medium. The results were the same when Na^+ was replaced by tetramethylammonium ions (TMA^+), by a mixture of Mg^{2+} and mannitol, or in most cells, by Tris. In several cells, however, 50% replacement of Na^+ by Tris produced a decrease in I_H. Since the other substituents never decreased I_H, we concluded that these cells were especially sensitive to some detrimental effect of Tris.

The Na^+ concentration dependences of I_D and I_H are plotted in Fig. 6. Over the range of 100–460 mM, I_D varies almost linearly with $[Na^+]_o$, suggesting that Na^+ is the major charge carrier for I_D. I_H is unaffected by changes in $[Na^+]_o$ in the range 50–460 mM, but drops rapidly for lower concentrations. It thus appears that Na^+ is

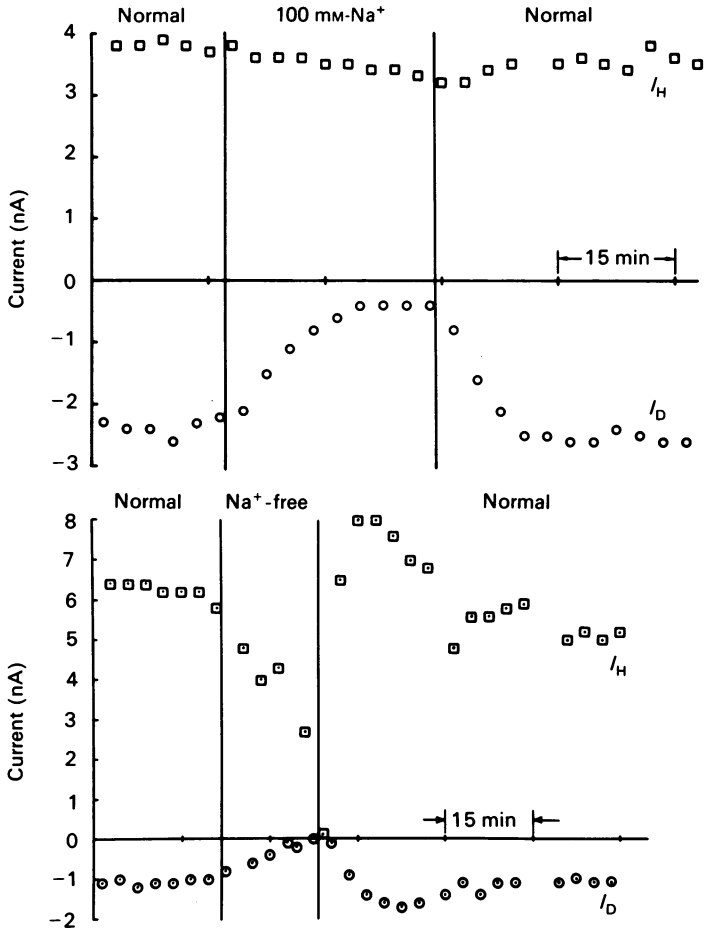


Fig. 5. Effect of reduction of $[Na^+]_o$ on I_D (○) and I_H (□). Top, $[Na^+]_o$ was reduced from 460 to 100 mM by substituting 360 mM-Tris. Bottom, reduction of $[Na^+]_o$ below 50 mM reduces I_H (⊡) in addition to I_D (⊙).

not a charge carrier for I_H and that the decrease in I_H at very low $[Na^+]_o$ is due to some secondary effect, perhaps an interference with Na^+–Ca^{2+} exchange. The effects of very low $[Na^+]_o$ are not due to inhibition of a ouabain-sensitive Na^+–K^+ pump, since both I_D and I_H persist for an hour or more after poisoning the pump with ouabain (W. B. Adams, unpublished results; Junge & Stephens, 1973, also report that bursting continues after ouabain poisoning).

K^+. Increases in extracellular K^+ concentration ($[K^+]_o$) and/or the application of K^+ channel blockers produced no decreases in either I_D or I_H. Fig. 7 illustrates the effects of increasing $[K^+]_o$ 4-fold. Assuming that the intracellular K^+ concentration ($[K^+]_i$) was unchanged during perfusion of high-K^+ medium, V_K (the K^+ equilibrium potential) should have been shifted from its value in normal medium of −75 mV (Kunze, Walker & Brown, 1971) to −40 mV. Since the holding potential for measurement of I_D and I_H was also −40 mV, the driving force for K^+ currents should

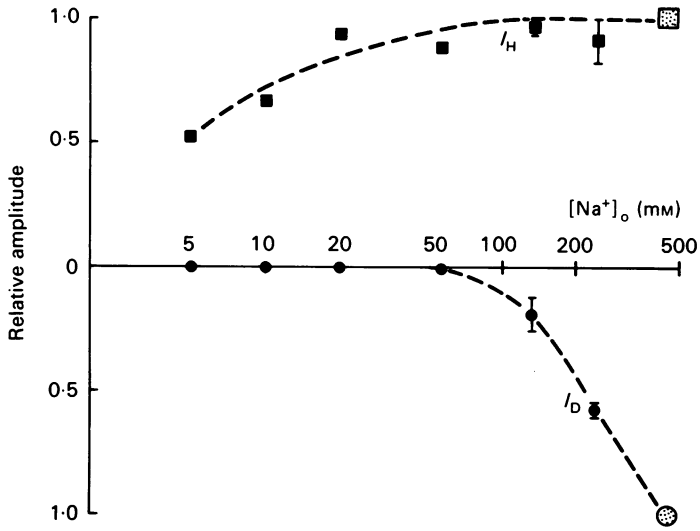


Fig. 6. $[Na^+]_o$ dependence of I_D and I_H. Pooled data from five experiments. I_D (●) varies nearly linearly with $[Na^+]_o$ down to 100 mM, while I_H (■) is essentially independent of $[Na^+]_o$ as long as $[Na^+]_o > 50$ mM. The data were normalized before pooling to give an amplitude in normal medium of 1·0 (stippled symbols).

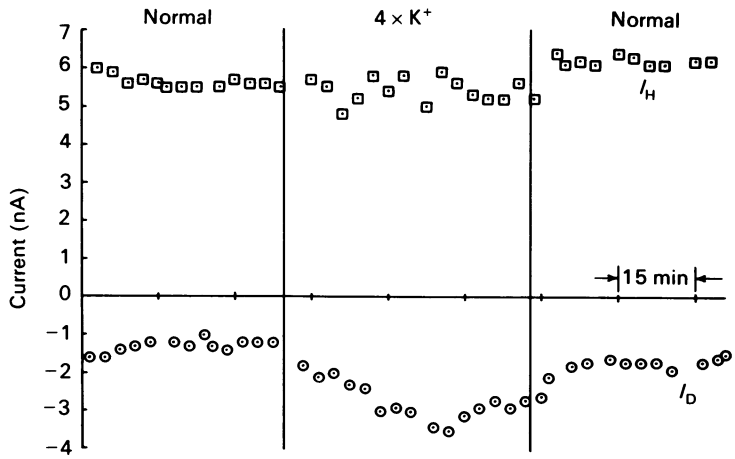


Fig. 7. Effect of increasing $[K^+]_o$ on I_D and I_H. $[K^+]_o$ was increased hyperosmotically from 10 to 40 mM (the osmolarity of normal medium is *ca.* 1200 mosmol). I_D (⊙) is increased while I_H (⊡) is unaffected.

have been zero or, allowing for small changes in $[K^+]_i$, very close to zero. As can be seen from Fig. 7, I_D was increased approximately 2-fold, while I_H was unchanged. The increase in I_D may result from an unmasking effect. In normal medium, the fast outward K^+ tail current immediately following an action potential or a depolarizing pulse appears partially to overlap I_D. Thus, when this tail current is reduced, I_D appears larger.

K^+ concentrations as high as 100 mM were perfused with qualitatively the same results. With such high concentrations, however, spontaneous synaptic input to R15

perturbed the base-line holding current to such an extent that quantitatively stable measurements were not possible. Nevertheless, I_H remained an outward current at −40 mV, even though a K^+ concentration of 100 mM in the bathing medium should have shifted V_K to a value 20–25 mV more positive than the holding potential.

Similar results were obtained with K^+ channel blockers. Cs^+ (up to 5 mM), 4-AP

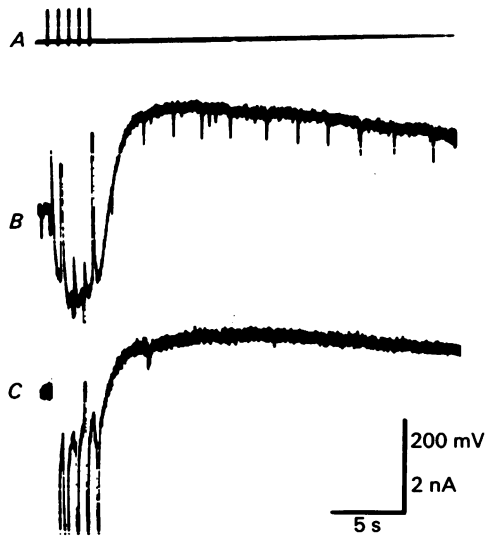


Fig. 8. Normal I_D and I_H in the presence of high $[K^+]_o$, TEA, Cs^+ and 4-AP. *A*, voltage. *B*, current from five depolarizing pulses in normal medium. *C*, $[K^+]_o$ was increased to 40 mM. In addition, TEA (50 mM) was added to block the delayed K^+ channels and the Ca^{2+}-activated K^+ channels, Cs^+ (2·5 mM) to block the anomalously rectifying K^+ channels, and 4-AP (5 mM) to block the K^+ A-current. The holding current exhibited large spontaneous fluctuations and the record shown was taken during a 10 s period when the holding current was relatively stable. When the voltage clamp was released, the cell fired 'action potentials' with durations between 100 and 1000 ms.

(4-aminopyridine, to 5 mM), and TEA (to 100 mM) were all without effect on both I_D and I_H (Fig. 8), even though we observed effective blocking of the resting K^+ conductance, the fast K^+ current (I_A), the delayed K^+ current (I_K), and the Ca^{2+}-activated K^+ current (I_C). Half-maximal blockage for the various K^+ currents have been reported as: resting K^+ current, 1 mM-Cs^+ (J. A. Benson, personal communication); I_A, 1·5 mM-4-AP (Thompson, 1977); I_K, 6 mM-TEA (Hermann & Gorman, 1979); I_C, 0·4 mM-TEA (Hermann & Gorman, 1979). With some of the blockers (notably TEA) I_D increased somewhat due to its being unmasked from the fast K^+ tail current that follows the depolarizing pulse, while I_H was unaffected. With very high concentrations of K^+ channel blockers, the increased size of spontaneous synaptic inputs again made it impossible to obtain quantitative measurements of I_D and I_H. (Under such conditions, when R15 was unclamped it fired action potentials with durations of up to 1000 ms.) Qualitatively, however, I_D and I_H appeared much the same, as illustrated in Fig. 8.

Ca^{2+}. Reductions in external Ca^{2+} brought about decreases in both I_D and I_H.

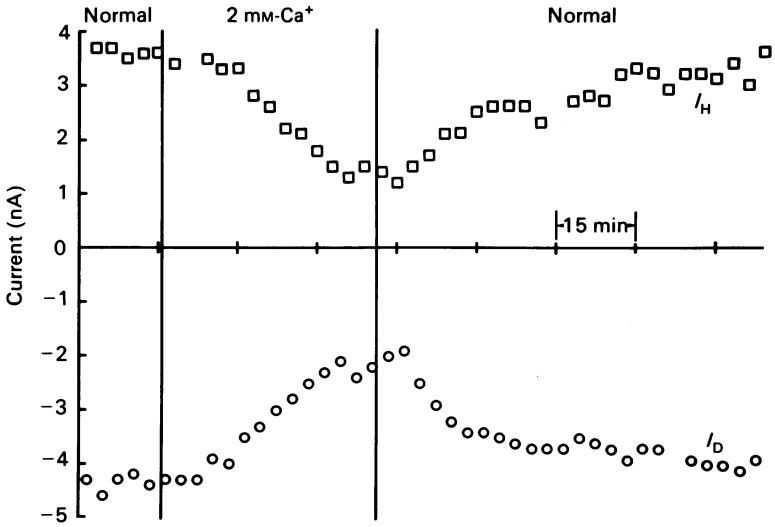


Fig. 9. Effect of reduction of $[Ca^{2+}]_o$ on I_D and I_H. $[Ca^{2+}]_o$ was reduced hyposmotically from 11 mM (normal medium) to 2 mM and then returned to normal medium. Both I_D (o) and I_H (□) were reduced in low Ca^{2+} medium.

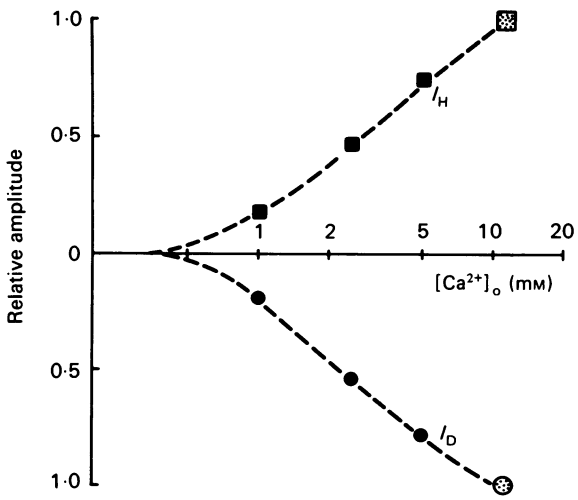


Fig. 10. $[Ca^{2+}]_o$ dependence of I_D and I_H. Data from a single experiment, expressed with respect to their amplitudes in normal medium (11 mM-Ca^{2+}, stippled symbols). Results from other experiments were similar, but showed shifts along the Ca^{2+}-concentration axis. Half-reduction of I_D (●) and I_H (■) was obtained in different experiments with $[Ca^{2+}]_o$ between 1·0 and 2·5 mM.

Fig. 9 illustrates the results of changing $[Ca^{2+}]_o$ from 11 to 2 mM and back again to normal medium. The Ca^{2+} concentration dependence for a single experiment is plotted in Fig. 10. I_D and I_H were reduced to half their amplitudes in normal medium by reducing $[Ca^{2+}]_o$ to approximately 2·5 mM. There appeared to be a great deal of individual variation in Ca^{2+} dependence (or perhaps seasonal variation, but this was

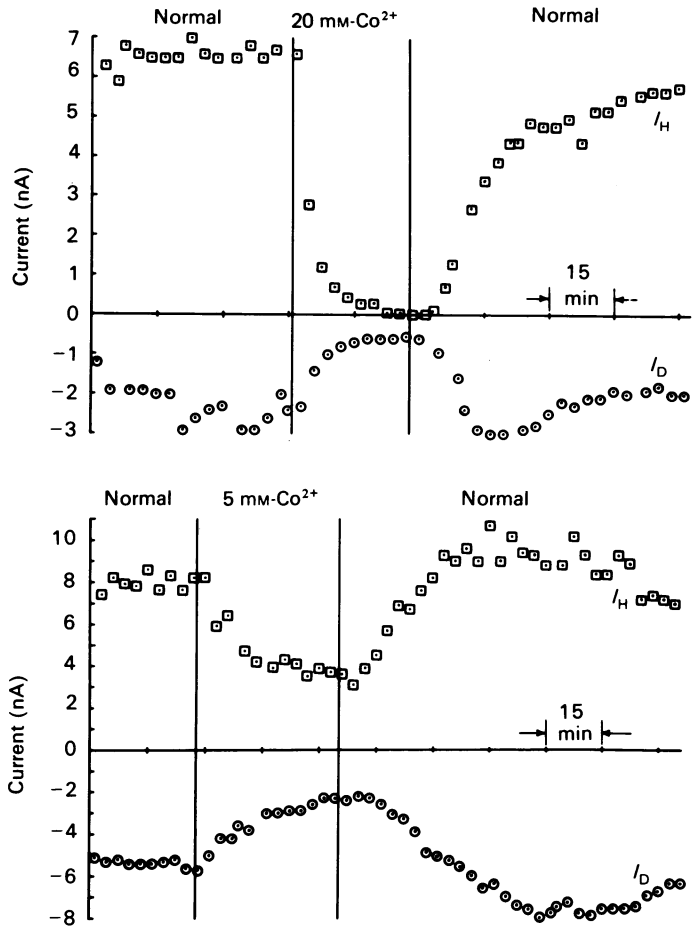


Fig. 11. Effect of Co^{2+} on I_D (●) and I_H (□). This Ca^{2+} channel blocker was added hyperosmotically at concentrations of 20 mM (top) or 5 mM (bottom). Qualititatively similar results were obtained with other blockers (Cd^{2+}, Ni^{2+}, see text).

not investigated sufficiently). In every cell tested, reduction of $[Ca^{2+}]_o$ to less than 0·5 mM reduced both I_D and I_H to less than 10% of their amplitudes in normal medium. The range of Ca^{2+} concentrations for which I_D and I_H were reduced to half-normal amplitude was 1–2·5 mM. In three cells (out of twelve tested), reduction of $[Ca^{2+}]_o$ to 5 or 5·5 mM produced an increase in I_D, but further reductions produced the usual blocking. Increasing $[Ca^{2+}]_o$ above its normal concentration also produced variable results, ranging from small increases in I_H (but not I_D) to reductions in both I_D and I_H.

The only ion that we found capable of supporting I_D and I_H in the absence of Ca^{2+} was Sr^{2+}. When $[Ca^{2+}]_o$ was replaced completely by Sr^{2+}, both I_D and I_H became much larger, and within a few minutes large oscillations in holding current (up to 100 nA in amplitude) were observed. These oscillations appeared to arise from bursting activity originating in poorly clamped regions of the axon. Partial substitution of Sr^{2+}

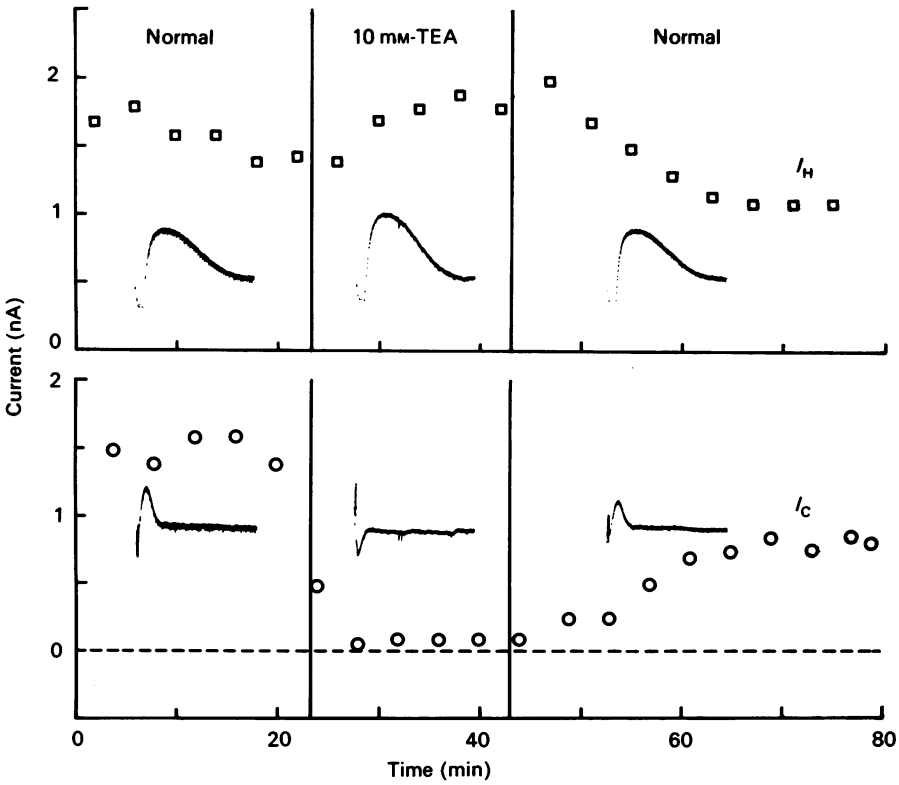


Fig. 12. Comparison of TEA sensitivity of I_H (□) and the Ca^{2+}-dependent K^+ current (○). I_H was elicited every 180 s by a train of five depolarizing pulses. Alternately, and 90 s separated from the I_H measurements, the Ca^{2+}-dependent K^+ current was activated by a 1 s 200 nA ionophoretic injection of Ca^{2+}. Successful injections were verified by monitoring the voltage-clamp holding current, which mirrored the ionophoretic current. TEA was added hyperosmotically at 10 mM. The Ca^{2+}-activated K^+ current was nearly totally blocked, revealing a Ca^{2+}-activated inward current (Kramer & Zucker, 1985*a*), while I_H increased slightly. Insets: sample current records.

for Ca^{2+} (Ca^{2+} 5·5 mM, Sr^{2+} 5·5 mM) increased I_D and I_H by several-fold, but again oscillations in the holding current made quantitative measurement impossible. When Ba^{2+} or Mg^{2+} were substituted for Ca^{2+}, both I_D and I_H were abolished.

The usual Ca^{2+} channel blockers (Co^{2+}, Cd^{2+}, Ni^{2+} and La^{3+}) blocked both I_D and I_H. Fig. 11 shows the result of adding 20 mM- and 5 mM-Co^{2+} hyperosmotically to the perfusion medium. Blockage occurred within 30 min and recovery required 30–60 min, depending on the concentration of the blocker. Half-maximal blockage was obtained with 5 mM-Co^{2+} (Fig. 11, lower half). The most effective blocker tried was Cd^{2+}, requiring only 50 μM for half-maximal blockage. With Ni^{2+}, half-maximal blockage was obtained with 3 mM, but the times required, both to initiate the block and for recovery, were much longer than with either Co^{2+} or Cd^{2+}. Blockage was also obtained with 1 mM-La^{3+}. However, the effects of La^{3+} were never completely reversible, so little use was made of this blocking agent.

Comparison of the effects of TEA on I_C and I_H. The evidence presented so far contains

no suggestion that K^+ is the charge carrier for the transient slow outward current. I_H, which is responsible for terminating the burst and for maintaining the interburst hyperpolarization, does not reverse at V_K (Figs. 1, 3 and 4), and is unaffected by changes in $[K^+]_o$ (Fig. 7) or by K^+ channel blockers (Fig. 8). There are, however, so many statements in the literature that the interburst hyperpolarization is mediated by a Ca^{2+}-activated K^+ conductance (I_C), that a direct comparison of I_C and I_H seemed advisable. To this end, experiments were performed in which measurements of I_C and I_H were made alternately before, during, and after perfusion with TEA. I_C was elicited by ionophoretic intracellular injection of Ca^{2+}; I_H was elicited by a train of five depolarizing pulses. The results of one such experiment are depicted in Fig. 12. I_C was found in preliminary experiments to be very sensitive to TEA, in accord with the 0·4 mM half-blocking effect reported by Hermann & Gorman (1979). In this experiment, 10 mM-TEA blocked I_C evoked by Ca^{2+} injection within 5 min of application (Fig. 12, bottom). No blockage of I_H was observed during the 20 min of perfusion with TEA (Fig. 12, top). I_H was, in fact, slightly increased by TEA. Upon removal of TEA from the perfusion medium, I_C recovered, albeit not to its original level. This lack of complete recovery may have reflected a leak of Ca^{2+} from the ionophoretic electrode or a movement of its intracellular location and a consequent change in the effectiveness of the injected Ca^{2+} that actually reached the membrane (cf. Gorman & Hermann, 1979; Tillotson & Gorman, 1980).

Currents in the presence of K^+ channel blockers

In normal medium, the early phase of I_D is obscured by the K^+ tail current that follows the depolarizing pulse (cf. Fig. 7). Since it was found that both I_D and I_H remained intact in the presence of elevated $[K^+]_o$ and of K^+ channel blockers, several experiments (Figs. 13–16) were carried out under these conditions to minimize K^+ currents during and after the depolarizing pulse, and thus unmask the inward current which underlies I_D.

Threshold-like behaviour. In the presence of K^+ channel blockers, inward tail currents were observed immediately following brief depolarizing pulses (Fig. 13). The tail currents consisted of at least two components, one of which decayed in 10 ms or less, and one of which lasted for hundreds of milliseconds. With 'subthreshold' pulses (10–20 mV) only the very fast tail current was seen; this component can probably be attributed to charging currents for the axon and the somal membrane. The amplitude of this fast charging current continued to grow with increasing pulse amplitude, but because of its rapid decay it does not contribute to I_D. With a 25 mV pulse, however, the much longer-lasting current which does seem to underlie I_D appeared. With larger pulses this slow component of the tail current assumed a nearly constant wave form, characterized by several inflexions and a non-exponential decay (see also Fig. 16*A* for an extended time scale). In Ca^{2+} or Na^+ plus Ca^{2+} medium, the current overshot zero at about 500 ms and remained positive for many seconds (Fig. 16*A*); in Na^+ medium, however, the current merely decayed monotonically to the base-line holding current (data not shown).

The threshold-like behaviour seen in the tail currents in Fig. 13 was investigated further in another R15 neurone with two purposes in mind: to look for abrupt changes in current during the pulse that corresponded to the abrupt activation of the tail

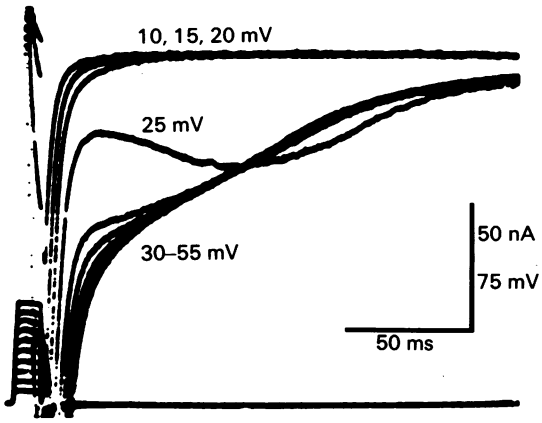


Fig. 13. Growth of the tail current in high (40 mM) $[K^+]_o$ and 100 mM-TEA. The amplitude of the depolarizing pulse was increased in 5 mV steps from 10 to 55 mV. With small pulses (10–20 mV), the tail current consisted primarily of the charging currents for the axon and the somal membrane capacitance. At 25 mV a much longer-lasting current appeared. The prominent inward current peak at approximately 75 ms after the 25 mV pulse was not apparent in the tail currents elicited by larger pulses, perhaps because it occurred closer to, or within, the pulse. With the larger pulses, the long-lasting tail current took on a constant wave form, while the charging currents immediately following the pulse continued to increase. The long-lasting tail current does not decrease smoothly, but has a number of inflexions.

current; and to ascertain the relationship between the inward tail current and the inward current seen in normal medium as I_D. The depolarizing pulse amplitude was increased in 1–2 mV increments and the current was measured at three times: (a) just before the end of the pulse (Fig. 14*A*); (b) at 100 ms after the pulse (Fig. 14*B*), sufficiently late that the simple axon charging currents should have disappeared (Fig. 13 and Connor, 1977); and (c) at 400 ms after the pulse (Fig. 14*C*), the time at which I_D typically reaches a peak in normal medium.

In Ca^{2+} medium, the current at the end of the pulse (Fig. 14*A*) increased smoothly up to 64 mV, and then decreased as the Ca^{2+} current peak occurred earlier in the pulse. (The decrease was probably due in part to Ca^{2+} inactivation and in part to the appearance of an outward K^+ current as its voltage-dependent TEA blocking became less effective.) The tail currents increased from practically no activation to full activation between 31 and 33 mV. In Na^+ plus Ca^{2+} medium the Na^+ current became activated at lower voltages than the Ca^{2+} current and the peak of the Na^+ current appeared just at the end of a 30 mV pulse. With larger pulses, the peak occurred earlier in the pulse and the current at the end of the pulse consisted primarily of the Ca^{2+} current. The tail currents measured at both 100 (Fig. 14*B*) and 400 ms (Fig. 14*C*) became fully activated between 24 and 25 mV, just in the range of potentials where the Na^+ current was beginning to activate.

Is the 'tail current' a tail current? In order to test whether this multiply inflected current was activated by the depolarization during the pulse or by the return of the membrane potential to the holding voltage, we compared the currents activated by pulses of identical amplitudes but differing durations. Fig. 15 shows the currents

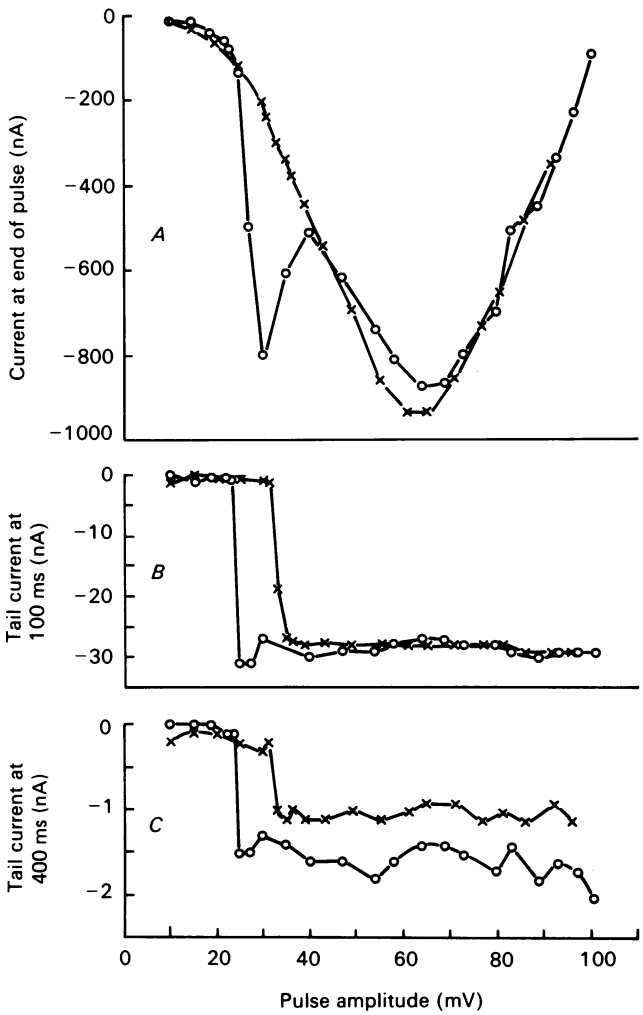


Fig. 14. Threshold-like activation of the inward tail current measured in the presence of high (40 mM) $[K^+]_o$ and TEA (100 mM). Holding potential −40 mV. The current was measured just before the end of a 10 ms depolarizing pulse (*A*), and at 100 (*B*) and 400 ms (*C*) after the pulse. ×: Ca^{2+} (low Na^+) medium, $[Na^+]_o$ 50 mM, $[Ca^{2+}]_o$ 11 mM. ○: Na^+ plus Ca^{2+} medium, $[Na^+]_o$ 460 mM, $[Ca^{2+}]_o$ 11 mM.

activated by 10 and 100 ms pulses. The sustained depolarization during the longer pulse (trace *b*) activated a sustained inward current and, hence, the current trace (*d*) lies below the tail-current trace (*c*) elicited by the shorter pulse (trace *a*). However, superimposed on the sustained current during the longer pulse is a transient current that mirrors in every detail the shape of the tail currents from the shorter pulse. Thus, activation of this transient inward current with its multiple inflexions requires only the initial depolarization for triggering and, once triggered, it runs its course even when the depolarization is maintained.

Blocking by a hyperpolarizing pulse. The threshold behaviour of the tail current

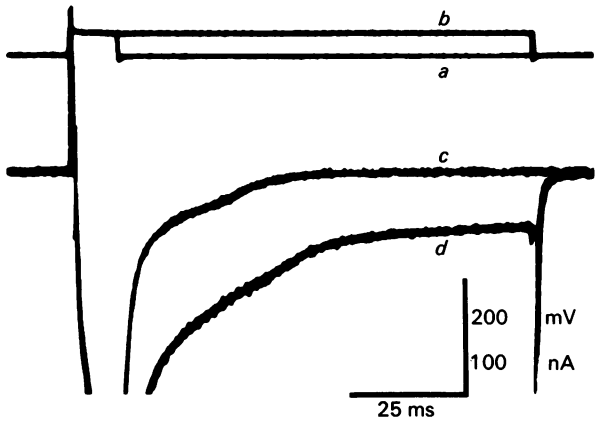


Fig. 15. The 'tail current' is not really a tail current. It is not necessary to return the membrane voltage to the holding potential in order to observe the long-lasting current that is seen as a 'tail current' with shorter pulses. When the pulse is extended from 10 ms (*a*) to 100 ms (*b*), the current seen following the 10 ms pulse (*c*) is visible in the current trace from the longer pulse (*d*), superimposed on a sustained inward current (the sustained inward current is sensitive to Cd^{2+} (data not shown) and thus is presumably carried by Ca^{2+}).

(Fig. 14), its complex wave form (Fig. 13), and the triggering phenomenon (Fig. 15) suggested that the depolarizing pulse activates an action potential in some unclamped or poorly clamped region of the axon, and that as the action potential propagates away from the soma, its effects are seen in the soma as an inward tail current. An alternative explanation, however, is that the influx of Na^+ and Ca^{2+} during the pulse triggers some mechanism in or near the soma that produces the tail current. In order to test this possibility, we presented 10 ms depolarizing pulses of sufficient amplitude to activate the tail current and then followed them immediately with 10 ms hyperpolarizing pulses before returning to the holding potential. The result of this procedure should be to increase the total Na^+ and Ca^{2+} influx, since channels opened by the depolarizing pulse will remain open for some time into the hyperpolarizing pulse and both Na^+ and Ca^{2+} will see an increased electrochemical gradient for influx into the cell. However, in Na^+ medium or in Ca^{2+} medium, we found that the normal tail current (Fig. 16*A*, trace *a*) was blocked completely with this procedure (trace *b*). The necessary amplitude of the hyperpolarizing pulse varied with the amplitude of the depolarizing pulse, from 4 mV for a 48 mV depolarizing pulse to 98 mV for a 119 mV depolarizing pulse (Fig. 16*B*). This finding supports the idea that the depolarizing pulse triggers an action potential at some point remote from the soma and that following the depolarizing pulse closely with a hyperpolarizing pulse can prevent threshold from being reached. With Na^+ plus Ca^{2+} medium, we have not been able to block the tail current with a single hyperpolarizing pulse, possibly because the kinetics of the Na^+ and Ca^{2+} channels are so different.

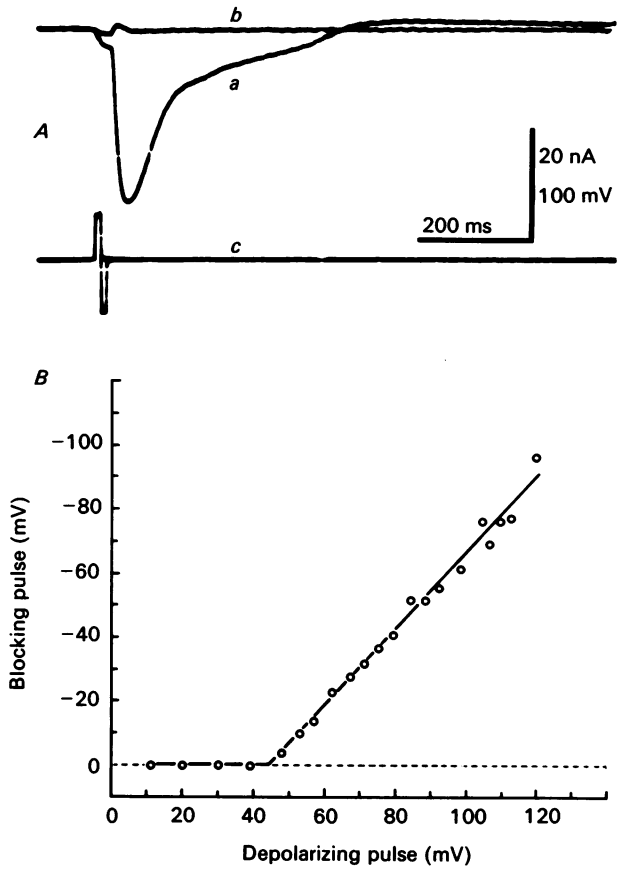


Fig. 16. The tail current can be blocked by following the depolarizing pulse with a hyperpolarizing pulse. *A*, in Ca^{2+} medium (50 mM-Na^{+}, 11 mM-Ca^{2+}) or in Na^{+} medium (460 mM-Na^{+}, 11 mM-Ca^{2+}, 100 μM-Cd^{2+}; data not shown), the tail current elicited by a 10 ms depolarizing pulse (trace *a*) can be blocked by following the pulse immediately by a 10 ms hyperpolarizing pulse of appropriate amplitude (trace *b*). The current signal was passed through a sample–hold amplifier which was placed in the 'hold' mode for 25 ms to eliminate the large current flows *during* the pulse. The initial current deflexions in traces *a* and *b* reflect an offset in the sample–hold amplifier. The two voltage traces, with and without a hyperpolarizing pulse, are shown superimposed in trace *c*. *B*, the necessary amplitude for the blocking pulse depends on the amplitude of the depolarizing pulse. In the absence of a hyperpolarizing pulse, depolarizing pulses of amplitude greater than 48 mV elicited tail currents with amplitudes that were independent of the amplitude of the depolarizing pulse (cf. Fig. 14). However, the amplitude of the hyperpolarizing pulse (ordinate) that was necessary to block the tail current depended almost linearly on the amplitude of the depolarizing pulse (abscissa).

DISCUSSION

The previous paper (Adams, 1985) described the kinetics of two slow tail currents in *Aplysia* neurone R15 that are activated by action potentials, and demonstrated that these two currents play an important role in the generation of bursting pace-maker activity in this cell. In this paper, we have attempted to determine the charge carriers for the two currents by investigating the ways in which they are dependent on

TABLE 1. Summary of the properties of I_D and I_H

Properties of I_D

1. Threshold-like activation.
2. An inward current between -20 and -120 mV.
3. Decreased by decreasing $[Na^+]_o$.
4. Decreased by decreasing $[Ca^{2+}]_o$ or by addition of Ca^{2+} channel blockers.
5. Unaffected (or increased) by increasing $[K^+]_o$ or by addition of K^+ channel blockers.

Properties of I_H

1. Graded activation.
2. An outward current between -20 and -120 mV.
3. Decreased by decreasing $[Ca^{2+}]_o$ or by addition of Ca^{2+} channel blockers.
4. Unaffected by decreasing $[Na^+]_o$ (as long as $[Na^+]_o > 50$ mM).
5. Unaffected by increasing $[K^+]_o$ or by addition of K^+ channel blockers.

membrane potential and on the ionic composition of the bathing medium. Table 1 summarizes the properties of I_D and I_H that may be relevant to a discussion of the possible charge carriers for these two currents. In brief, I_D has a steep activation curve and is Na^+ and Ca^{2+} dependent while I_H shows a graded activation and is only Ca^{2+} dependent. Over the range of membrane potentials between -20 and -120 mV, I_D is seen as an inward current and I_H as an outward current.

I_D

The tail currents recorded following reduction of K^+ currents (Fig. 13) suggest a quite different time course for I_D than was apparent in normal medium. Under these conditions, the tail current reaches its full amplitude within a few milliseconds following the depolarizing pulse and then decays monotonically, although not smoothly, back to the base line. Although the procedures used to block K^+ currents may well change the shape and amplitude of the tail current in some quantitative aspects, it seems likely that a qualitatively similar tail current is present in normal medium, but is obscured by the K^+ current activated during the depolarizing pulse.

Probably the strongest evidence linking I_D to action potential generation is provided by its threshold-like activation. When I_D is elicited in normal medium, full activation is obtained with an increase in the height of the depolarizing pulse of as much as 10 mV in some cells (e.g. Fig. 2), and as little as 2–3 mV in other cells. Activation of I_D is not entirely all-or-none, as pulses close to the apparent threshold can elicit inward tail currents of less than full amplitude both in normal medium (Fig. 2 and Lewis, 1984) and even to some extent after blocking K^+ channels (Fig. 13). Nevertheless, the activation curve for I_D is much steeper than that for I_H (Fig. 2). Further support for the threshold hypothesis is provided by the dual pulse experiment illustrated in Fig. 16. The finding that larger depolarizing pulses require larger hyperpolarizing pulses to block activation of the tail current (Fig. 16*B*), even though the size of the tail current does not increase with depolarizing pulse amplitude (Fig. 14), argues strongly against the possibility that the depolarizing pulse activates some process that is then inactivated by the hyperpolarizing pulse. Instead it seems more likely that the net polarization from the two pulses is summed, and if the sum exceeds threshold, then the tail current is generated.

The question remains, however, if I_D is the result of action potential generation, where is the action potential generated? A number of pieces of evidence point to a location somewhere in the axon away from the soma. First, the action potential does not arise in the soma. The soma remains well clamped during the depolarizing pulse, as seen by the voltage recording electrode and, in several experiments, by a third electrode placed in the cell to monitor the effectiveness of the clamp. Secondly, the location must be far enough distant from the soma so that the summation time constant in the dual pulse experiment is at least 20 ms. Finally, in a number of experiments in which the axon was ligatured, or in which the soma was undercut and removed from the ganglion, we were unable to detect any trace of I_D. Kramer & Zucker (1985*a*) report also that their phase II current, which can be measured in left upper quadrant cells after axotomy, and in R15 before axotomy, disappears in R15 when this cell is axotomized. It is possible, of course, that R15 is simply more sensitive to damage from axotomy. However, when taken in conjunction with the difference in activation curves (i.e. I_D activates with a threshold-like dependence on pulse potential; phase II current in the left upper quadrant cells activates gradually with pulse potential), it appears more likely that there is a real difference in the way these currents are generated in the two groups of cells.

Although our evidence indicates that the appearance of I_D is dependent upon the generation of an action potential in the axon, we still do not know the connexion between the action potential and the current detected in the soma as I_D. The somal currents may arise simply from a passive spread of current into the soma as the action potential propagates away, or the axonal action potential may activate a separate set of conductances through with I_D flows. In either case, it would appear that I_D is a mixed Na^+–Ca^{2+} current, since it can be supported by either ion when K^+ currents are blocked. The dual requirement for *both* Na^+ and Ca^{2+} in normal medium may well reflect the dual requirement for Na^+ and Ca^{2+} for action potential generation in R15 (Carpenter & Gunn, 1970). Such an interpretation would be consistent with the more relaxed requirement for *either* Na^+ or Ca^{2+} after reduction of I_K. The voltage dependences shown in Figs. 1, 3 and 4 are applicable in any event, since they are measured *following* the occurrence of the action potential. However, the activation curves in Figs. 2 and 14 might not measure directly the activation of the conductances for I_D, but only the activation, or triggering, of the action potential. Certainly it is important to point out that I_D is not simply a voltage-clamp artifact. Although it is seen in our experiments because the axon cannot be well clamped by electrodes in the soma, it also will be generated by action potentials propagating along the axon in normally bursting cells.

I_H

Probably the clearest indication of the charge carrier for I_H is given by its sensitivity to changes in $[Ca^{2+}]_o$ and to Ca^{2+} channel blockers and its insensitivity to changes in $[Na^+]_o$ or $[K^+]_o$ or to K^+ channel blockers. Without speculating about the mechanisms underlying activation of the current, these observations point to Ca^{2+} as the charge carrier for I_H. The lack of a reversal potential for I_H anywhere between -20 and -120 mV also seems to rule out K^+ as a charge carrier for I_H. Lewis (1984) has also noted the lack of reversal of a slow hyperpolarizing after-current in neurone

R15, and points out that this will have to be explained before it can be asserted that it is carried by K^+. However, we recognize the possible difficulty of measuring in the soma a reversal potential for a conductance change that may occur in the axon. Thus, while we regard the lack of a reversal potential as corroborative evidence for our conclusion that K^+ is not the charge carrier, we consider the strongest evidence to be given by the ion changes and pharmacological treatments.

If Ca^{2+} is indeed the charge carrier for I_H, how can Ca^{2+} flow produce an outward current when the Ca^{2+} equilibrium potential is much more positive than the potentials at which I_H is measured? There seems only one answer to that question, that there is a resting inward Ca^{2+} current (Eckert & Lux, 1976; Gorman *et al.* 1982) and that the outward current seen as I_H results in fact from an inhibition of this resting Ca^{2+} current. The inhibition must, in turn, be produced by some aspect of the cell's activity during the burst or, more particularly, since I_H can be elicited by single action potentials, by some process occurring during action potential generation. One possibility for the nature of the process is simply the membrane potential excursions during the action potential. However, there seems to be little voltage-dependent inactivation or inhibition of Ca^{2+} currents (Eckert & Ewald, 1983). A second possibility is that it is the accumulation of some ion as the result of the ion fluxes during action potential generation. Here, the supportive evidence is strong. Tillotson (1979) and Eckert & Tillotson (1981) have demonstrated Ca^{2+}-dependent inhibition of Ca^{2+} channels in R15 and other *Aplysia* neurones, and a similar demonstration has been given by Standen (1981) in *Helix* cells. Moreover, the Ca^{2+} influx during a single 15 ms depolarizing pulse is sufficient to partially inhibit Ca^{2+} conductance (Eckert & Tillotson, 1981), and the recovery from inhibition follows a two- (or more) exponential time course (Tillotson & Horn, 1978; Plant & Standen, 1981) with time constants that are similar to those found in the preceding paper for the decay of I_H (Adams, 1985, Fig. 8). Thus, although our own experiments provide no direct evidence about the underlying mechanisms, our tentative conclusion is that I_H results from a Ca^{2+}-dependent inactivation of a resting Ca^{2+} current.

It appears that R15 is not unique in possessing a mechanism that causes a resting Ca^{2+} conductance to decrease as a result of action potential activity. Based on evidence similar to ours, Kramer & Zucker (1985*b*) have concluded that the interburst hyperpolarization phase of axotomized left upper quadrant cells in the *Aplysia* abdominal ganglion is also mediated by an action-potential-dependent reduction in a resting Ca^{2+} current. On the other hand, this mechanism is not a universal one, even among bursting cells. In cells of the guinea-pig hippocampus, the current that mediates the interburst hyperpolarization has been shown to have a K^+-dependent reversal potential and, thus, is most likely mediated by a K^+ current (Brown & Griffith, 1983).

The evidence presented above should probably be sufficient to establish Ca^{2+} as the charge carrier for I_H. However, in bursting cells in general, and in R15 in particular, it has so long been accepted that 'the interburst hyperpolarization is mediated by a Ca^{2+}-activated K^+ conductance' that it behoves us to review the differences between the experiments that led to that conclusion and the experiments reported in this paper. Because of the difficulties in interpreting non-voltage-clamp data obtained from a cell with so many voltage-dependent conductances, we will

concentrate largely on evidence obtained from experiments under voltage clamp. Generally, those findings fall into four categories:

(*1*) *Changes in membrane slope conductance measured during the interburst phase.* When hyperpolarizing current pulses are injected during the interburst, the resulting membrane voltage deflexions are smallest just after the end of the burst and gradually increase throughout the interburst period (Junge & Stephens, 1973), as though a

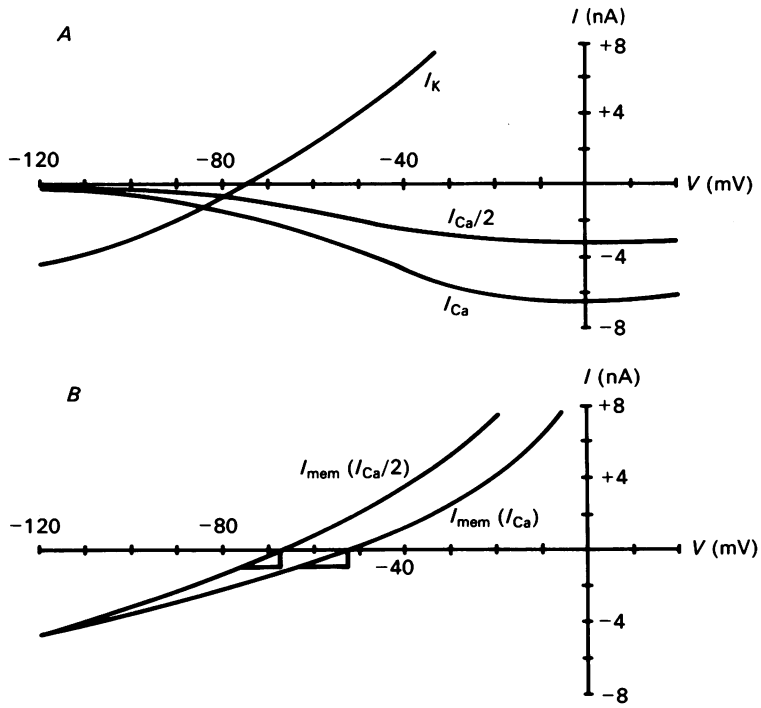


Fig. 17. Illustration of how an increase in membrane slope conductance can be caused by a decrease in an inward current. The total membrane current (*B*: I_{mem}) is constructed from two components (*A*), a K^+ current (I_K) and a Ca^{2+} current (I_{Ca}). I_K intersects the abscissa at the K^+ equilibrium potential (−75 mV) and has an upward curvature. The magnitude of I_{Ca} increases with depolarization; its voltage dependence has been modelled after that described by Gorman *et al.* (1982). Reduction of I_{Ca} by half (*A*: $I_{Ca}/2$) leads to an increase in the slope of the membrane *I–V* curve (*B*: $I_{mem}(I_{Ca}/2)$). Following the traditional electrophysiological method, injection of a 1 nA current would produce a voltage deflexion of 11 mV with the $I_{mem}(I_{Ca})$ curve and 8 mV with the $I_{mem}(I_{Ca}/2)$ curve, corresponding to a 37 % increase in membrane slope conductance (B: ⌐┘).

conductance were activated by the burst and then slowly turned off. Our interpretation, however, is that I_H arises from a Ca^{2+} conductance that is turned *off* during the burst and then slowly turns back *on* during the interburst. The conflict in this case is more apparent than real. In the range of membrane potentials over which these conductance measurements are made, the Ca^{2+} conductance is highly voltage dependent, decreasing rapidly with hyperpolarization (cf. Gorman *et al.* 1982). As a result, the Ca^{2+} current becomes smaller, or less negative. Another way of describing this relationship is to

say that in this voltage range, the Ca^{2+} current contributes a negative slope conductance (Fig. 17*A*: I_{Ca}). Since the total slope conductance of the membrane at any particular voltage (Fig. 17*B*: I_{mem}) is the sum of all the slope conductances that are active at that voltage, a *decrease* in the size of a negative slope conductance (Fig. 17*A*: $I_{Ca}/2$) will appear as an *increase* in total membrane slope conductance (Fig. 17*B*: $I_{mem}(I_{Ca}/2)$; see also Adams, Parnas & Levitan, 1980).

(*2*) *Activation by depolarizing pulses of an outward current that varies inversely in amplitude with* $[K^+]_o$. Johnston (1980) and Carnevale & Wachtel (1980) found that long (1–12 s) depolarizing pulses activated Ca^{2+}-dependent outward tail currents. These currents increased in amplitude with decreasing $[K^+]_o$, and extrapolation of the *I–V* curves gave 'reversal potentials' near the K^+ equilibrium potential. However, in none of these experiments was it reported possible to reverse these slow currents. And as we have shown (see Results, following Figs. 1, 3 and 4), extrapolation of the *I–V* curves for I_H would indeed predict a reversal potential somewhere near the K^+ equilibrium potential even though measurement of I_H over a wider range of membrane potentials shows it not to reverse at all. Moreover, in at least some of those experiments, the cause for the K^+-dependent increase in outward current amplitude appears to have been secondarily, rather than directly, related to the changes in $[K^+]_o$. Examination of the reported recordings (e.g. Fig. 9, Carnevale & Wachtel, 1980) reveals that in some cases when $[K^+]_o$ was decreased, a larger number of axonal action potentials were triggered during the depolarizing pulse. This would have, in itself, caused an increase in I_H. Finally, the slow outward current described in these reports is 'relatively insensitive' to TEA (Johnston, 1980), which is at odds with the high TEA sensitivity of the Ca^{2+}-activated K^+ conductance (Hermann & Gorman, 1979; see also Fig. 12, this paper), but in accord with our interpretation of I_H as a decrease in Ca^{2+} current.

(*3*) *Changes in extracellular* K^+ *accumulation during the burst cycle found using* K^+*-sensitive electrodes.* By placing K^+-sensitive electrodes near the outer surface of R15, Gorman *et al.* (1982) measured changes during the burst cycle of local K^+ concentration, which presumably resulted from changes in K^+ efflux during the cycle. However, the major source of K^+ efflux comes from the action potentials during the burst, and the authors of this study conclude that their 'results indicate that much of the change in $[K^+]_o$ during the post-burst hyperpolarization represents the diffusion of K^+ from the membrane to the extracellular K^+-sensitive electrode and suggest that there may be very little movement of K^+ [across the membrane] during this period'.

(*4*) *Demonstration of a* Ca^{2+}*-activated* K^+ *conductance in R15.* Perhaps the most difficult argument to counter, simply because it is the most general, is that the presence of a Ca^{2+}-activated K^+ current (I_C) has been demonstrated in R15 by injection of Ca^{2+} (Meech, 1972; Gorman & Hermann, 1979). Furthermore, aequorin and Arsenazo III experiments have shown that intracellular Ca^{2+} concentration ($[Ca^{2+}]_i$) increases during the burst, as a result of the Ca^{2+} that flows in with each action potential (Stinnakre & Tauc, 1973; Gorman & Thomas, 1978; Smith & Zucker, 1980). Thus it seems that I_C should be gradually activated during the burst, in much the same way as we have described for activation of I_H. And yet, we see no indication that any long-lasting K^+ current is activated, either in normally bursting cells (e.g.

the lack of a reversal potential for I_H in Fig. 1) or following a train of depolarizing pulses (e.g. the lack of reversal potential and insensitivity to TEA and increased $[K^+]_o$ in Figs. 3, 4, 7, 8 and 12). In this respect our findings are complemented by the TEA insensitivity and absence of reversal potential for the slow outward current in R15 (Johnston, 1980; Carnevale & Wachtel, 1980) and for the phase III current in left upper quadrant neurones (Kramer & Zucker, 1985*a*, *b*). To be sure, we can activate a long-lasting K^+ current by injecting Ca^{2+} (Fig. 12), using amounts similar to those reported elsewhere (Gorman & Thomas, 1980). But we have never seen activation of such a long-lasting current as a result of the cell's electrical activity. So the question becomes: why is a long-lasting Ca^{2+}-dependent K^+ current *not* activated by action potentials and short depolarizing pulses? The only answer that appears plausible is that the amount of Ca^{2+} that enters the cell during action potential generation is not sufficient to activate a long-lasting I_C.

We believe that our interpretation of I_H as resulting from a decrease in a Ca^{2+} current that is active at rest provides the most consistent explanation for the variety of experimental data now available on bursting in neurone R15. This explanation encompasses not only the observed dependence on $[Ca^{2+}]_o$ and the changes in membrane conductance during the interburst period, but also provides answers for the lack of reversal of this current at the K^+ equilibrium potential and for its insensitivity to K^+ channel blockers. Although we have not studied directly the manner by which the Ca^{2+} current is decreased, it seems possible that Ca^{2+}-induced inactivation of Ca^{2+} channels (Eckert, Tillotson & Brehm, 1981) may provide the mechanism.

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